Practical –11

Aim: Write a program for Floyd-Warshal algorithm.

Algorithm:

input : No. of vertices and an adjacency matrix
output : A matrix showing the shortest path b/w every pair of vertices

⇒ start
⇒ Read the number of vertices and the weighted matrix
⇒ initialize a distance matrix equal to the input matrix
⇒ set an outer loop for intermediate vertex k for 0 to V-1
⇒ set two inner loops for source i and destination j from 0 to V-1
⇒ update paths using formula:

dist[i][j] = min(dist[i][k], dist[k][j])

⇒ print the final shortest distance matrix
⇒ stop.

Program:

```
#include <iostream>

using namespace std;


int main() {

   int n;

   int dist[10][10];
```

```
    cout << "Enter number of vertices: ";

cin >> n;

    cout << "Enter the adjacency matrix:\n";
    cout << "(Enter 99999 if there is no direct edge)\n";

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            cin >> dist[i][j];
        }
    }

    // Floyd-Warshall Algorithm
    for (int k = 0; k < n; k++) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (dist[i][k] != 99999 &&
                    dist[k][j] != 99999) {

                    if (dist[i][k] + dist[k][j] < dist[i][j]) {
                        dist[i][j] =
                            dist[i][k] + dist[k][j];
                    }
                }
```

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```
            }
        }
    }
    cout << "\nShortest Distance Matrix:\n";

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (dist[i][j] == 99999)
                cout << "INF\t";
            else
                cout << dist[i][j] << "\t";
        }
        cout << endl;
    }

    return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter the adjacency matrix:
(Enter 99999 if there is no direct edge)
0 5 99999 10
99999 0 3 99999
99999 99999 0 1
99999 99999 99999 0

Shortest Distance Matrix:
0    5    8    9
INF  0    3    4
INF  INF  0    1
INF  INF  INF  0
```

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Time complexity :

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```
for(int k=0 ; k<n; k++)  -> n times
    {  for (int i=0; i<n;i++) -> n time
        {  for (int j=0 ;j<n;j++) -> ntime.
           { if (dist[i][k]+dist[k][j] <
                 dist[i][j])
              {
                dist[i][j] = dist[i][k]+
                             dist[k][j];
              }
           }
        }
    }
```

Total TC = O(n x n x n)
= O(n^3)

Best case = O(n^3)
Average case = O(n^3)
worst case = O(n^3)

=>The algorithm uses three nested loops each ,executing n time ,regardless of the graphs edge weights.

Conclusion :

Thus, we have successfully implemented the Floyd–Warshall Algorithm in C++ to find the shortest paths between all pairs of vertices in a weighted graph. The algorithm updates the distance matrix by considering each vertex as an intermediate vertex. The best-case, average-case, and worst-case time complexities are all \(O(n^3)\).

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